

DSC 209 Project 3

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1 Introduction

Within this project, we explore how interactive visualizations can facilitate players and analysts in better understanding the dynamics between team balance and hero performance within the multiplayer online battle area game *Mobile Legends: Bang Bang*.

As the *Bang Bang Mobile Gang*, our goal was to bridge the raw hero statistics with a player-center design that was both intuitive and efficient. Inspired by the *Airline On-Time* dataset, where we created a meaningful visualization to transform complex, multidimensional data into actionable insights, we wanted to apply similar principles to game analytics. The diversity within this dataset including numerical statistics, categorical player roles, and performance data, also requires thoughtful visual encoding to reveal significant relationships.

In other words, we wanted to create a tool to facilitate exploratory analysis with our guiding question: 'How can we visualize hero performance to encourage informed and balanced team selection?' Through visualization, we enable users to explore how hero attributes, win rates, and release timelines may interact with one another to shape a competitive balance and effective composition.

2 Preprocessing

Our dataset originates from a public Kaggle dataset featuring *Mobile Legends: Bang Bang* hero statistics. This includes information on hero attributes, roles, and professional performance data. Some attributes were chosen to be excluded due to containing large amounts of missing or irrelevant data.

Data Cleaning Process

We standardized inconsistent data formats within the `Release.Date` column, bridging the multiple varying representations into a uniform structure. Moreover, we computed additional derived features, such as overall win rate, to provide users with an interpretable and comparative performance metric. Lastly, numerical attributes were normalized to a $[0, 1]$ range scale to allow differing hero types to be fairly compared on the radar chart.

3 Development

The development of our visualization was an iterative and collaborative process that required consistent coordination, implementation, technical experimentation, and reflective design choices. We utilized `D3.js` and `JavaScript` to create our visualization, with the aid of `HTML` and `CSS` for structure and styling. The radar chart allows us to highlight how heroes across multiple core statistics can differ.

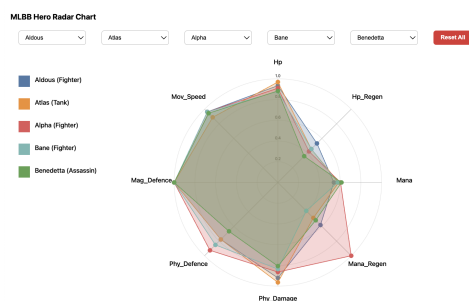
Our team divided responsibilities to optimize efficiency and leverage individual strengths. Jason spearheaded the search for a dataset that was both engaging and suitable for an exploratory visualization assignment to set the foundation for our analytical direction. Following this, Laura oversaw the repository management and established a shared asynchronous workspace to create a structured and transparent work space. Baixi focused on developing and refining the initial working models iteratively improving and making pivotal efforts to improve the visual and interactive components in response to the team's feedback and requests. Pooja played a key role in quality assurance and project consistency by reviewing visual outputs, identifying inconsistencies, and consistently reevaluating to maintain

steady progress and workflow. Finally, Jason, Pooja, and Laura led the documentation and formal write-up to ensure that the design rationale and decisions were clearly articulated.

In total, we approached this project incrementally, dedicating approximately twelve hours to documentation and formal writing, alongside an additional period for data preprocessing, coding, visualization development, and refining.

4 Visualization

Hero Stat Profile Radar Chart



The radar chart visualizes the identity of each hero utilizing an eight-dimensional representation of their attributes. The octagonal display allows players to compare multiple heroes across numerous key statistics to provide a comprehensive view of their strengths and roles. More specifically, each of the axes corresponds to a core attribute including HP, Moving Speed, Mag Defence, Physical Defence, Physical Damage, Mana Regen, Mana, and HP Regen.

Users can select up to five heroes via drop-down menus, hover over each chosen character for additional tool-tip details and statistics, and reset their selection. Each axis is normalized to ensure proper comparability and enable meaningful assessment of overall hero balance.

From a design standpoint, we were inspired by the visual representations used in sports analytics and Pokémon statistic charts. These served as ideal templates to make complex numerical data intuitively readable at a glance. Additionally, we explored alternative visualizations, such as scatterplot and trellis plots, to examine hero performance, strengths, and weaknesses. However, we determined that these designs were less effective in achieving our goal. These approaches risked

splitting attention across multiple visual panels and introducing unnecessary cognitive load. Instead, the radar chart enhanced the versatile nature of hero identity, achieving a balance between comprehensiveness and clarity to support our goal of providing players with a data-rich tool for informed analysis.

Ultimately, we found it essential to clarify the nature of the data underlying our visualization (refer to the addendum for further details). From a quick case study, we performed several key transformations on the dataset, including the computation of derived attributes such as win rate and release year. In particular, the inclusion of win rate is significant in providing a performance-based metric when evaluating heroes. More specifically, it was important to select a hero with a win-rate of 40-50% range to raise insightful questions about decision-making and optimization: what justifications exist for choosing a statistically weaker option when seeking to maximize team success?

5 Conclusion and Summary

This project reflects the collaborative effort and diverse skill sets of our team. We thoroughly transformed a complex, multidimensional dataset into an interactive and accessible visualization tool to empower players to explore strategy and balance within the game.

We view this work as a technical exercise and step toward making data-driven insights approachable for competitive gaming communities.

6 Addendum: Project Notes and Reflections

Throughout development, we maintained a log of design deliberations and troubleshooting notes. These include the different stages from initial brainstorming to the implementation of interactivity, color schemes, and performance tuning.

Link: Addendum